

Week 1 — Introduction to Blockchain

Based on Lecture 01 and Lecture 02.

1. What this course is actually about

The slides make one thing very clear: **blockchain does not always mean Bitcoin**. The course is not mainly about “how to buy crypto” or “how to invest.” Instead, it is about the underlying technology behind blockchain systems, their properties, weaknesses, smart contracts, and financial uses like stablecoins and tokenization.

Easy understanding

A lot of people hear “blockchain” and immediately think:

- Bitcoin
- price charts
- trading
- making money

But this course is saying: no, first understand the **technology**.

Main focus of the course

The course will teach:

- technologies used to build blockchains
- properties and pitfalls of blockchains
- smart contracts
- cryptocurrencies as one use case
- financial applications like stablecoins and tokenization

What the course will not teach

It will **not** teach:

- how to open a crypto account
- investment strategies
- what regulations should be adopted

So in the midterm, if sir asks “What is the scope of this course?” or “Why blockchain does not simply mean Bitcoin?” this is the idea.

2. Bitcoin statistics — why were they shown?

The lecture starts with Bitcoin statistics such as coins in circulation, total supply of 21 million, blocks mined, blockchain size, transactions per second, and the fact that Bitcoin handles only around 7 transactions per second while Visa handles around 24k. It also says Bitcoin is only one cryptocurrency among many.

Why these numbers matter

These statistics are not random. They are there to show:

Point 1: Bitcoin is huge but limited

Bitcoin has a massive market cap and global presence, but it still has technical limitations such as low transaction throughput. So being famous does not mean perfect.

Point 2: Blockchain is broader than Bitcoin

Even though Bitcoin is the most famous example, it is only one use case in a much bigger ecosystem.

Point 3: Scale and design tradeoffs matter

A blockchain may be secure and decentralized, but still be slower or heavier than traditional systems. That is why blockchain design choices matter.

3. The cryptocurrency avalanche

The slides mention 18,000+ cryptocurrencies, hundreds listed on major exchanges, hundreds of millions of users, many exchanges, and many businesses accepting crypto. But the course still says the real focus is **blockchain technology** and more broadly **distributed ledger technology (DLT)**, whose concepts are older than crypto itself.

Easy explanation

This means:

- crypto became popular very fast
- many coins exist

- many may fail
- the important academic thing is the technology underneath

Important midterm takeaway

If asked “Why does the course focus on blockchain rather than crypto hype?” answer:

- because cryptocurrencies are only one application
 - blockchain ideas existed before modern crypto popularity
 - DLT has wider uses beyond digital coins
-

4. Why was blockchain created?

This is one of the most important topics from Week 1.

The lecture says blockchain was motivated by problems in digital payments:

- dependence on intermediaries
- intermediaries can block or reverse payments
- intermediaries may not always be online
- double-spending problem
- lack of financial freedom
- global transactions being slow and expensive

Easy explanation of the motivation

Before blockchain, online payments usually needed a **middleman**, like a bank, company, or payment service.

Problem 1: Intermediaries

If a bank is in the middle:

- it can stop a payment
- reverse a payment
- delay a payment
- charge fees

That means you do not have full control.

Problem 2: Double-spending

This is a classic digital money problem.

If money is purely digital, what stops someone from copying the same digital money and sending it to two different people?

With physical cash, this is hard. If I give you one real note, I no longer have it. But with digital information, copying is easy. So we need a system that ensures the same money is not spent twice.

Problem 3: Financial freedom

The slide references the 2008 bailouts. The point is that people wanted a system where financial activity does not depend completely on central institutions.

Problem 4: Global payments are slow and costly

Traditional international transfers can be:

- slow
- expensive
- dependent on several intermediaries

Blockchain tried to offer a more direct peer-to-peer model.

Big idea

Blockchain was created to enable **direct P2P digital payments without relying fully on a trusted central party**.

5. What is blockchain?

The lecture defines blockchain as a **ledger for recording data**, especially transactions, across a network of computers called **nodes**. Some nodes keep part of the ledger, and some keep a full ledger. Transactions can be initiated by nodes, end users, or organizations such as exchanges.

Easy definition

A blockchain is basically:

- a record book

- shared across many computers
- used to record transactions and other data
- updated without one single owner controlling everything

Ledger meaning

A ledger is like an account book or transaction record.

Distributed ledger meaning

A distributed ledger means that the record is not stored only in one place. It is spread across many computers in the network.

Nodes

A node is simply a computer running blockchain software.

Some nodes:

- store only part of the ledger
- store the full ledger
- validate and share transactions

Very simple real-life analogy

Think of a classroom where every student keeps a notebook of class payments. If everyone updates their notebook and compares with others, that becomes a distributed record instead of one notebook held by one monitor.

6. Why is it called a blockchain?

Because transactions are grouped into **blocks**, and those blocks are linked together using **cryptographic hashes**.

Structure

A blockchain is called a blockchain because:

- data is stored in blocks
- each block refers to the previous one
- the linking happens through hashes

This creates a chain of blocks.

7. Cryptographic hashing

This is another core topic.

The lecture says a cryptographic hash function:

- takes input text of any length
- produces a fixed-length random-looking output
- even a tiny change in input causes a major change in output
- can be computed by standard algorithms like SHA-256

Easy explanation

A hash function is like a machine:

- you put in data
- it gives you a code-like output
- same input always gives same output
- tiny input change gives very different output

Example idea

“Hello” and “Hello2” will produce completely different hashes.

Why fixed length matters

Even if the input is long or short, the output size stays fixed.

Why random-looking matters

The output looks meaningless and unpredictable.

Why it helps blockchain

Because if someone changes block data, the hash changes too, so tampering becomes visible.

8. Chaining blocks using hashes

Each block contains the hash of the previous block. That is how blocks are linked together.

Easy explanation

Suppose:

- Block 56 contains the hash of Block 55
- Block 57 contains the hash of Block 56

Now imagine someone changes Block 55.

Then:

- Block 55's hash changes
- Block 56 still has the old previous hash
- so the chain no longer matches
- the tampering gets detected

That is why changing one old block affects later blocks too.

9. Practical blockchain hashing and NONCE

The slides say that in a practical blockchain, hashes must satisfy some criteria, for example starting with a certain number of zeros. If the block's hash does not satisfy the rule, the block is not accepted. Since transaction data is fixed, miners add a fictitious number called a **NONCE** and keep changing it until the resulting hash meets the required condition. This process is called **mining**.

Easy explanation

A blockchain does not accept just any hash.

It may require:

- hash starts with 000
- or meets some difficulty rule

But transaction data usually cannot be changed freely.

So what can be changed?

The **NONCE**.

NONCE

A nonce is an extra value added into the block.

The miner:

- tries one nonce
- computes the hash
- checks if it meets the condition
- if not, tries another nonce
- repeats until success

This trial-and-error process is mining.

Why this matters

Mining makes block creation difficult and costly. That difficulty helps secure the blockchain.

10. What does blockchain bring to the table?

According to the slides, blockchain offers:

- immutability / tamper resistance
- auditability
- decentralization

A. Immutability

This means data is very hard to change once added.

Why?

- if one transaction changes, the block hash changes
- then later blocks also become inconsistent

- recomputing all of that is very hard

It is not absolutely impossible in theory, but practically very difficult.

B. Auditability

All transactions are available and traceable.

That means:

- history stays recorded
- records can be checked
- actions are easier to verify

C. Decentralization

There is no single central authority that fully controls everything.

Benefits:

- more autonomy
- censorship resistance
- lower single point of failure

Important subtle point

The slides also raise “Privacy? Anonymity?” meaning blockchain does not automatically mean full privacy. That is an important thought for later topics.

11. Tampering and unchaining

The “Securing + Chaining of Blocks” slide shows that if data in a block is changed, the block’s hash changes, and the chain becomes broken or “unchained.” Any such change is easily detectable.

Easy understanding

Suppose a transaction was:

- Asim → Arif: \$20

and someone changes it to:

- Asim → Arif: \$50

That tiny transaction change changes the entire hash.

Then the next block still points to the old hash, so the chain no longer matches.

Result

Tampering is detected immediately by comparing hashes.

12. Can blockchain still be attacked?

The lecture says permanent tampering is possible, but would require **more than 50% control**.

Easy explanation

This refers to a majority attack idea.

If an attacker controls most of the network's power or influence, they may rewrite history more effectively than the honest network.

Why it matters

Blockchain is very secure, but not magic.

Its security depends on:

- honest majority
- cost of attack being high
- distributed participation

So if asked "Is blockchain absolutely impossible to attack?" the correct answer is no. It is highly resistant, not invincible.

13. Security of transactions

The slides raise important questions:

- Has the payer authorized the transaction?
- Is the payee the right person?
- Is someone spending more than they own?
- Can we verify this efficiently?

They also mention **digital signatures**, **public key cryptography**, and **simplified payment verification (SPV)** as concepts to study later.

Easy understanding

For a transaction to be accepted, the system needs to know:

- the sender really approved it
- the sender has enough balance
- the transaction is valid and not fake

Traditional banks do this centrally. Blockchain must do it in a distributed way.

14. When should blockchain be used?

The lecture says blockchain is useful when:

- many transactions are generated by multiple parties
- parties have a common interest
- central control should be avoided
- agreement is needed over shared data

Easy explanation

Blockchain is useful when many people need to share and trust a common record, but do not want one single party to control it.

Examples

- digital money transfer
- shared ownership records
- multi-party tracking systems
- ratings or shared data that everyone should agree on

Important takeaway

Blockchain is not for every problem. If one trusted party can manage a normal database, blockchain may be unnecessary.

15. Multi-party consensus

The lecture says this boils down to a **multi-party consensus system**, where everyone agrees on shared data such as:

- Muneeb transferred PKR 70k to Zain
- a restaurant rating is 3.1/5

Easy explanation

Consensus means:
multiple participants agree on one common version of truth.

Without consensus:

- one node may say payment happened
- another may say it did not
- the system becomes unreliable

With consensus:

- everyone converges to the same accepted record
-

16. Consensus options: PoW and PoS

The lecture introduces two major consensus options:

Proof of Work (PoW)

- block creators spend energy
- in return they get reward
- block creators are called miners

Proof of Stake (PoS)

- block creators lock up cryptocurrency
- in return they get fees
- block creators are called validators

Easy comparison

Proof of Work

In PoW, miners prove effort by doing computational work. Mining is expensive because it uses energy.

Proof of Stake

In PoS, validators prove commitment by staking coins rather than burning computation in the same way.

Core point

Both are methods to achieve consensus and decide who gets to add the next block.

17. Validity of a blockchain

The slides say a blockchain is legitimate when all its blocks are valid. A block is valid if:

- it has the correct hash of the previous block
- its own hash meets the required criteria in PoW
- all transactions inside it are valid

Easy explanation

A blockchain is not valid just because blocks exist.

It must satisfy all three layers:

- chain linkage is correct
- consensus rule is satisfied
- transactions are genuine

Transaction validity example

If Alice gives 0.3 BTC to Bob, the system must check:

- did Alice actually have it?
 - did Alice approve the payment?
 - is the transaction properly formed?
-

18. Centralized database vs distributed ledger

The lecture visually compares centralized databases and distributed ledgers. A centralized database has more centralized control, while a distributed ledger spreads data handling across multiple nodes and multiple entities.

Easy explanation

Centralized database

- one main authority controls it
- one place or one system is dominant
- easier to manage, but more dependent on trust and single-point control

Distributed ledger

- multiple participants maintain records
- no single party must fully control updates
- agreement is built through synchronization and consensus

Core difference

A centralized database is about one trusted owner.

A distributed ledger is about shared trust and coordination.

19. DLT: accurate description

This is very important and easy to confuse.

The slides explain:

- multiple nodes are distributed in a network
- transaction data is generated and shared by them
- each node keeps its own journal or version of the database
- the blockchain is the unified view of these individual journals
- the goal is to keep journals in sync most of the time
- the ledger is not the database itself; it is the mechanism of synchronizing different journals

Easy understanding

Journal

Each node keeps its own local record, like its own notebook.

Why journals can differ

They may differ because:

- messages arrive at different times
- a node is temporarily offline
- transactions are still being validated
- network delays exist

Blockchain as unified view

The blockchain is not one magical notebook floating in the air. It is the common agreed view that emerges when all these journals synchronize.

Super important exam line

The ledger is not simply the database; it is the synchronization mechanism and shared agreed view across distributed journals.

That line is very midterm-worthy.

20. Syncing and consensus are hard

The final Week 1 slide says:

- there is a set of journals
- each is an individual view of shared data
- the goal is convergence to one shared view
- the ledger emerges from synchronization
- distributed consensus is hard

Easy explanation

This is hard because:

- no central boss exists
- nodes may receive information at different times
- some nodes may fail or act maliciously
- everyone still has to agree

That difficulty is exactly why consensus algorithms are such a big topic in blockchain.

WEEK 1 QUESTION BANK WITH DETAILED ANSWERS

Now the important part.

I'll give you exam-style questions of different kinds.

Q1. Define blockchain in simple words.

Answer

1. Blockchain is a **distributed ledger** used to record data, especially transactions, across a network of computers called nodes.
 2. Instead of one central database owner, multiple nodes participate in storing and sharing the ledger.
 3. Data is grouped into blocks and linked together, creating a secure and traceable chain.
-

Q2. Why was blockchain created?

Answer

1. Blockchain was created to reduce reliance on intermediaries like banks that can delay, block, or reverse payments.
 2. It helps solve the **double-spending problem** in digital peer-to-peer payments.
 3. It also aims to support more direct, global, fast, and low-cost digital transactions with greater financial autonomy.
-

Q3. What is the double-spending problem?

Answer

1. Double-spending means trying to spend the same digital money more than once.
 2. This is difficult with physical cash, but easy to attempt in digital systems because digital data can be copied.
 3. Blockchain prevents this by maintaining a shared transaction history that the network agrees on.
-

Q4. Why is blockchain called a blockchain?

Answer

1. Data is stored in **blocks**, where each block contains a set of transactions.
 2. These blocks are linked to one another using the hash of the previous block.
 3. Because the blocks are cryptographically connected in sequence, the structure forms a **chain of blocks**.
-

Q5. What is a cryptographic hash function?

Answer

1. A cryptographic hash function takes input data of any length and produces a fixed-length output.
2. The output looks random, but the same input always gives the same hash.

3. Even a tiny change in the input causes a major change in the output, which helps detect tampering.
-

Q6. Write three important properties of hashing in blockchain.

Answer

1. It converts any size input into a fixed-length output.
 2. It is highly sensitive to input changes, so even one small modification changes the hash greatly.
 3. It helps maintain block integrity because tampered data produces a different hash.
-

Q7. Explain how blocks are chained together.

Answer

1. Each block stores the hash of the previous block.
 2. This previous-hash reference links one block to the earlier block in sequence.
 3. If an older block changes, its hash changes too, which breaks the link and makes tampering visible.
-

Q8. What is a nonce?

Answer

1. A nonce is an additional value inserted into a block.
 2. It is changed repeatedly by miners to produce a hash that meets the required criteria.
 3. Since transaction data is fixed, the nonce is the flexible field used during mining.
-

Q9. What is mining?

Answer

1. Mining is the process of repeatedly changing the nonce and recomputing the block hash.
 2. The goal is to find a hash that satisfies a required condition, such as starting with a number of zeros.
 3. In Proof of Work systems, successful miners get rewards for creating valid blocks.
-

Q10. What is meant by immutability in blockchain?

Answer

1. Immutability means that once data is added to the blockchain, it becomes very difficult to alter.
 2. If one block is changed, its hash changes, so later blocks must also be recomputed.
 3. This makes unauthorized modification extremely hard and helps protect transaction history.
-

Q11. What is auditability in blockchain?

Answer

1. Auditability means that transactions can be tracked and checked.
 2. Historical records remain available on the chain for verification.
 3. This improves transparency and makes it easier to examine past activity.
-

Q12. What is decentralization?

Answer

1. Decentralization means no single central authority fully controls the ledger.
 2. Multiple nodes participate in maintaining and validating the system.
 3. This reduces censorship risk and lowers dependence on a single point of failure.
-

Q13. What happens if a block is tampered with?

Answer

1. The data change causes the block's hash to change.
 2. The next block still contains the old previous hash, so the chain linkage breaks.
 3. As a result, tampering becomes detectable because the blocks no longer match correctly.
-

Q14. Is blockchain fully impossible to attack?

Answer

1. No, blockchain is not absolutely impossible to attack.
 2. The lecture notes that permanent tampering may be possible if an attacker gains more than 50% control.
 3. So blockchain offers very strong security, but it still depends on the network remaining mostly honest.
-

Q15. When should blockchain be used?

Answer

1. Blockchain is useful when many transactions are produced by multiple parties.
 2. It is especially useful when those parties share an interest but do not want one central controller.
 3. It works best when a system needs common agreement over shared data.
-

Q16. What is consensus in blockchain?

Answer

1. Consensus is the process by which network participants agree on one shared version of data.
 2. It ensures the blockchain becomes a single converged view instead of many conflicting copies.
 3. Without consensus, nodes could disagree on transactions and the ledger would not be reliable.
-

Q17. Differentiate between Proof of Work and Proof of Stake.

Answer

1. In Proof of Work, block creators spend computational power and energy; in Proof of Stake, they lock up cryptocurrency as stake.
 2. PoW block creators are called **miners**, while PoS block creators are called **validators**.
 3. PoW rewards work done through computation, while PoS selects participants based on staked assets and gives fees or rewards.
-

Q18. What makes a block valid?

Answer

1. A valid block contains the correct hash of the previous block.
 2. In a PoW blockchain, its own hash must meet the required difficulty criteria.
 3. All transactions inside the block must also be valid.
-

Q19. What makes a transaction valid?

Answer

1. The sender must actually have the funds being spent.
 2. The sender must have authorized the transaction.
 3. The transaction must be correctly addressed and accepted under the rules of the blockchain.
-

Q20. Centralized database vs distributed ledger.

Answer

1. A centralized database is controlled mainly by one authority, while a distributed ledger is shared across multiple nodes or entities.

2. In a distributed ledger, agreement must be reached without relying on one central controller.
 3. A centralized system is simpler to control, but a distributed ledger provides shared trust and reduces single-point dependence.
-

Q21. What is DLT?

Answer

1. DLT stands for **Distributed Ledger Technology**.
 2. In DLT, multiple nodes keep their own versions or journals of shared data.
 3. The goal is to keep these journals synchronized so that a common agreed view emerges.
-

Q22. Why can different journals be different in DLT?

Answer

1. Nodes may receive transactions at different times due to network delays.
 2. Some nodes may be temporarily offline or slower in updating.
 3. Validation and relay timing can differ, so local records may briefly not match until consensus is reached.
-

Q23. What is meant by saying “the ledger is not the database”?

Answer

1. In blockchain, the ledger is not just one stored database file.
 2. It is the shared agreed state that emerges from syncing many individual journals.
 3. So the ledger is better understood as a synchronization and agreement mechanism across nodes.
-

Q24. Why is distributed consensus hard?

Answer

1. There is no central authority telling everyone what the correct version is.
 2. Nodes may have different local views, delays, failures, or malicious behavior.
 3. Even with these challenges, the network must still converge to one accepted shared record.
-

NUMERICAL / LOGICAL / SIMULATION QUESTIONS

Q25. A blockchain creates 144 blocks per day and each block currently gives 3.125 BTC reward. How many new bitcoins are created per day?

Answer

1. Formula:
New bitcoins per day = blocks per day × reward per block
 2. Calculation:
 $144 \times 3.125 = 450 \text{ BTC}$
 3. Therefore, **450 new bitcoins per day** are produced at that reward rate. This exact example is also mentioned in the lecture.
-

Q26. If one transaction in Block 55 is changed, explain step by step what happens to Blocks 56 and 57.

Answer

1. Changing a transaction in Block 55 changes the contents of Block 55, so its hash becomes different.
2. Block 56 still contains the old previous hash of Block 55, so the link between Block 55 and Block 56 becomes invalid.

3. Because Block 56's contents would also need updating, the effect propagates forward and Block 57 also becomes inconsistent.
-

Q27. Suppose a miner tries nonce values 1, 2, 3, 4 and none produce a hash starting with 000. What should the miner do next?

Answer

1. The miner should continue trying new nonce values.
 2. The block data usually cannot be changed, so the nonce remains the adjustable field.
 3. This repeated trial-and-error process continues until a valid hash meeting the criteria is found.
-

DIAGRAM QUESTIONS

Q28. Draw and explain the basic structure of three chained blocks.

Answer

You can draw it like this:

Block 1: [Transactions | prev hash = 0 | nonce] ---> Hash H1
Block 2: [Transactions | prev hash = H1 | nonce] ---> Hash H2
Block 3: [Transactions | prev hash = H2 | nonce] ---> Hash H3

3-point explanation

1. Each block contains its own transactions plus the previous block's hash.
 2. This previous hash creates the chain relationship between blocks.
 3. If any earlier block changes, its hash changes and later links break.
-

Q29. Draw a simple diagram showing tampering.

Original:

Block 55 --H55--> Block 56 --H56--> Block 57

After tampering in Block 55:

Block 55* --H55'--> Block 56(still points to old H55) --> mismatch

Answer

1. Tampering changes the contents of Block 55, so its new hash becomes H55'.
 2. Block 56 still stores the earlier expected hash, so it no longer matches the changed block.
 3. This mismatch exposes tampering and breaks the continuity of the chain.
-

DIFFERENCE QUESTIONS

Q30. Differentiate between blockchain and Bitcoin.

Answer

1. Blockchain is the underlying technology, while Bitcoin is one application built on that technology.
 2. Blockchain can be used in many other systems beyond cryptocurrency.
 3. The course itself emphasizes that blockchain does not always mean Bitcoin.
-

Q31. Differentiate between a normal database and a distributed ledger.

Answer

1. A normal database is often centrally controlled, while a distributed ledger is maintained across multiple nodes.
2. A distributed ledger relies on synchronization and consensus rather than only one administrator.

3. Blockchain records are shared and agreed upon collectively, unlike conventional centralized storage.
-

Q32. Differentiate between a full ledger holder and a partial ledger holder.

Answer

1. A full ledger holder stores the entire blockchain history, while a partial holder stores only part of it.
 2. Full holders can independently verify more of the chain's contents.
 3. Partial holders help participate with lower storage requirements but depend more on summarized or limited data.
-

THEORETICAL LONG QUESTIONS

Q33. Explain the role of hashing in blockchain security.

Answer

1. Hashing gives each block a unique fixed-length fingerprint based on its contents.
 2. If even a small part of a block changes, its hash changes drastically, exposing tampering.
 3. Since each block stores the previous block's hash, hashing also helps link blocks together securely across the chain.
-

Q34. Explain why consensus is necessary in a blockchain system.

Answer

1. In a distributed system, different nodes may have different local views of transactions.
2. Consensus helps them converge to one shared accepted history, called the distributed ledger.

3. Without consensus, there would be no single trusted record of who paid whom or what data is correct.
-

Q35. Explain the statement: “Blockchain is the unified view of individual journals.”

Answer

1. Each node keeps its own journal or local record of transactions.
 2. These journals are synchronized by sharing, validating, and agreeing on transaction data.
 3. The blockchain is the final shared view that emerges when these journals converge closely enough through consensus.
-

VERY LIKELY MIDTERM SHORT QUESTIONS

These are the ones I think sir can very realistically ask from Week 1:

1. Define blockchain.
2. Why was blockchain created?
3. What is double-spending?
4. Why is blockchain called a blockchain?
5. What is a cryptographic hash?
6. What is a nonce?
7. What is mining?
8. What is immutability?
9. What is auditability?
10. What is decentralization?
11. What happens if block data is tampered with?
12. What is consensus and why is it needed?
13. Compare PoW and PoS.
14. When should blockchain be used?
15. Differentiate centralized database and distributed ledger.
16. Explain DLT using journals.
17. What makes a block valid?
18. Can blockchain be attacked?

SUPER SHORT MEMORY SHEET FOR WEEK 1

Use this for quick revision.

Blockchain

- distributed ledger
- records transactions
- shared across nodes

Why created

- remove intermediaries
- solve double-spending
- support direct digital P2P payments

Block structure

- transactions
- previous block hash
- nonce

Hashing

- fixed-length output
- tiny change in input = huge change in output
- used for integrity

Mining

- keep changing nonce
- aim: valid hash
- PoW block creation

Main benefits

- immutability
- auditability
- decentralization

Consensus

- shared agreement
- one converged ledger
- needed in distributed systems

DLT

- many journals
- syncing those journals
- ledger = shared agreed view